CNN logits on the PC and on the STM32N657

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The question

One MNIST CNN: two 3×3 convolutions, two pools, two linear layers.
Three ways to run it:

- PC, FP32 checkpoint.
- PC, every weight cast to IEEE FP16 and stored back in FP32.
- The NUCLEO-N657X0-Q, same rounding, Helium, FP32 activations.

A fourth program, Cube.AI's FP32 graph, was already measured on the same 200 images.

Three different scores

- Logit MSE: mean over images of $\|z - z_{\mathrm{FP32}}\|_2^2$. Ten pre-softmax numbers.
- Cosine: mean over images of the cosine between those two vectors.
- Label accuracy: fraction of images whose largest logit is the written digit.

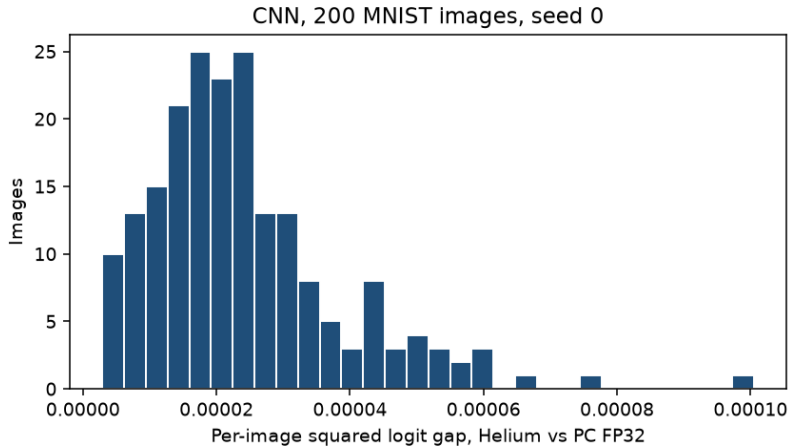
The subset is seed 0, 200 test images, pixels divided by 255.

What matches

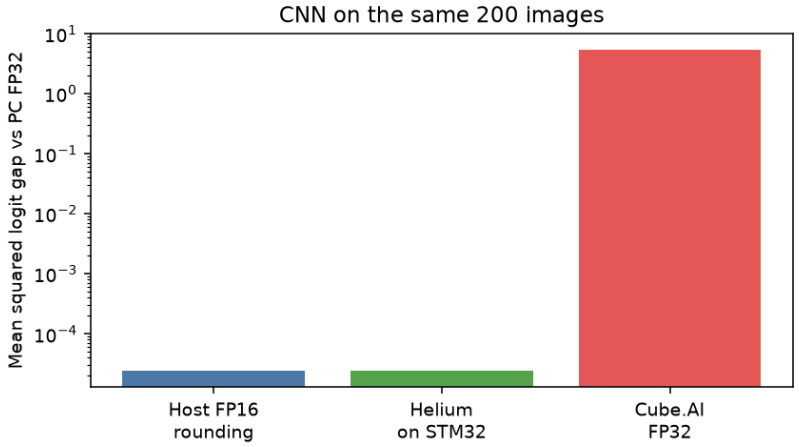
Run	Logit MSE	Cosine	Class differs
PC, FP16 rounding	2.422×10^{-5}	1.000	0/200
Helium on the board	2.422×10^{-5}	1.000	0/200
Cube.AI FP32	5.445	0.998	0/200

Label accuracy is 0.995 for the FP32 network and for all three rows. 199 of 200 digits.

Helium stays next to the PC



Cube.AI FP32 does not

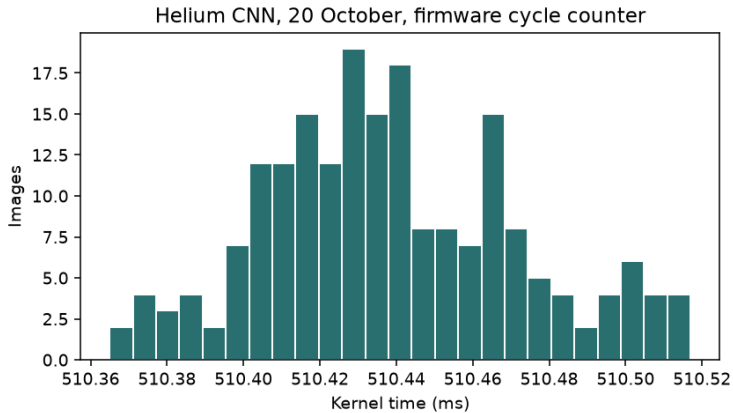


The right-hand bar is a different compiled graph: 11 software steps and 4 accelerator steps. The class still matches.

Same images, logarithmic gaps



Time on the board



Mean kernel time 510.4 ms on the 20 October run. The serial transfer is not included.

The same CNN, scored with `round_fp16`

The checkpoint was trained for 10 epochs (best MNIST test accuracy 99.25%). The Helium firmware on the board is that network: FP16 weights, FP32 activations. On the PC, `stm32nn.round_fp16` cast every parameter. `stm32nn.compare` then scored 200 images, seed 0.

- Host, after `round_fp16`: logit MSE 2.422×10^{-5}, class changes 0/200.
- Board, this morning: logit MSE 2.422×10^{-5}, class changes 0/200, 510.4 ms.

The graded operation

`stm32nn.round_fp16` is one line of PyTorch:

```
x.to(torch.float16).to(torch.float32)
```

Each finite float32 element is rounded to IEEE binary16, ties to even, and stored as float32. An integer tensor raises `RuntimeError` from `TORCH_CHECK`.

What stm32nn adds

A C++ PyTorch module on the PC. It does not flash the board.

- `round_fp16`: the cast used for the matching Helium run.
- `to_nhwc`, `to_nchw`: channel-last buffers, as in the generated CNN.
- `quantize_per_channel`: symmetric int8, scale $= \max|w|/127$.
- `conv2d`, `linear`, `max_pool2d`: separate steps, optional ReLU.
- `compare`: MSE, class disagreement, cosine.
- `check_export`: FP16 weights, a large prime channel count, pixels not divided by 255.

What the extension does not do

- It does not replace Cube.AI's convolution.
- The multiply-add loop Cube.AI calls is inside a Cortex-M55 library, not in a C file we can copy.
- A custom ONNX operator is not compiled by ST Edge AI.
- The 5.445 gap remains until that generated graph is changed, or until the PC calls that same graph.

Tonight

The probe enumerated, the serial port answered, and the Helium CNN ran all 200 images.
Result: the board and the PC rounding agree. Cube.AI's FP32 CNN, on the same images, does not agree in the logits and does agree in the class.